Effects of Weight Quantization on Turkish Morphosyntactic Performance in EuroLLM

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Abstract

Weight quantization reduces the resource requirements of large language models, but the preservation of specific grammatical contrasts requires targeted evaluation. This study compares the local FP16 reference of the multilingual EuroLLM-1.7B base model with MLX affine 8-bit and 4-bit weight quantization, using a group size of 64. The evaluation comprises 1,000 deduplicated minimal pairs sampled equally from four TurBLiMP categories: subject agreement, argument structure, nominalization, and irregular forms. The primary score is mean token log probability, with a beginning-of-sequence context and no end-of-sequence token. FP16, Q8, and Q4 achieve accuracies of 91.2%, 91.5%, and 88.4%, respectively. The Q8 difference of +0.3 percentage points is not significant (two-sided exact McNemar test, Holm-adjusted p = .25). The Q4 difference of −2.8 percentage points is significant (adjusted p = .000469; conservative paired 95% confidence interval [−5.151, −0.416]). Q4 produces 42 correct-to-incorrect and 14 incorrect-to-correct transitions; 46 transitions occur in the lowest quartile of absolute FP16 margins. Exploratory fragmentation analyses do not support a monotonically increasing relationship between the token-to-word ratio and decision changes. Measured peak MLX allocator memory is 63.8% lower for Q4 than FP16. The results establish a model- and sample-specific accuracy–memory trade-off, without demonstrating Q8 equivalence, general Turkish proficiency, or a causal fragmentation effect.

Keywords: weight quantization, Turkish, morphosyntax, minimal pairs, subword fragmentation.

1. Introduction

Running language models on local hardware requires sufficient memory for both the weights and the computations performed during inference. Low-precision methods can reduce inference memory requirements, but their numerical designs differ. LLM.int8() combines 8-bit multiplication with a higher-precision path for outlier features, whereas SmoothQuant transforms weights and activations to support W8A8 inference (Dettmers et al., 2022; Xiao et al., 2023). These efficiency results motivate a separate empirical question: whether quantization preserves a model's preferences on specific grammatical contrasts.

In Turkish, grammatical relations such as agreement, case selection, and subordination are often expressed through morphology. Tokenization is therefore relevant to how linguistic information is presented to a model. Toraman et al. (2023) compared five tokenization approaches, including a morphological-level tokenizer, in Turkish language models evaluated on six downstream tasks. Their

study motivates attention to tokenization choices, but it does not establish how a fixed tokenizer interacts with weight quantization in minimal-pair judgments. A decline in accuracy alone does not identify the grammatical or representational cause.

We evaluate three precision configurations of EuroLLM-1.7B (Martins et al., 2024) on the same Turkish minimal pairs from TurBLiMP (Başar et al., 2025). We ask how much 8-bit and 4-bit quantization change accuracy relative to FP16, where decision changes occur across categories and baseline margins, and whether the token-to-word ratio is associated with Q4 decision changes after accounting for category, sentence length, and baseline margin. Measurements of scoring time and memory use place the linguistic results in the context of local inference.

2. Related Work

2.1. Low-precision inference

Quantization schemes with the same nominal bit width can implement different numerical operations. LLM.int8() uses vector-wise quantization for most features and a 16-bit computation path for outlier dimensions (Dettmers et al., 2022). SmoothQuant uses a mathematically equivalent transformation to shift quantization difficulty from activations to weights, enabling 8-bit quantization of both (Xiao et al., 2023). These methods illustrate why bit width alone does not specify a quantization procedure. The present comparison holds the MLX affine scheme and group size fixed and varies the weight bit width. No further training or task-specific fine-tuning was performed.

2.2. Efficiency and Turkish tokenization

Published efficiency gains depend on the evaluated model, numerical method, and workload. Dettmers et al. (2022) report memory savings for mixed-precision 8-bit inference, while Xiao et al. (2023) evaluate memory and speed for W8A8 inference. Neither result establishes preservation of Turkish morphosyntactic preferences under MLX affine weight quantization. For Turkish, Toraman et al. (2023) show that tokenization granularity and vocabulary size warrant empirical evaluation across downstream tasks. Our study connects these concerns by measuring grammatical preferences and local resource use while holding the tokenizer fixed. Differences in tasks and quantization procedures preclude direct numerical rankings across these studies.

2.3. Minimal pairs and morphology

BLiMP uses minimally contrasting sentences to probe language models' sensitivity to grammatical distinctions in English (Warstadt et al., 2020). TurBLiMP provides the Turkish minimal pairs used in the present experiment (Başar et al., 2025). Turkish tokenization research compares character-, subword-, word-, and morphology-related representations, with evaluations on downstream tasks rather than the quantized minimal-pair task studied here (Toraman et al., 2023). We treat minimal-

pair accuracy as sensitivity to the selected contrasts. Our token-to-word ratio is a sentence-level fragmentation proxy; it does not identify morphemes or measure alignment with gold morphological boundaries.

3. Theoretical Framework

3.1. Weight approximation and preference margins

Groupwise affine quantization represents weights approximately using low-bit integer codes and a scale and offset shared within each group (MLX Contributors, n.d.). Approximation error can alter the probabilities assigned to sentence tokens without changing the preferred sentence. Conversely, a small shift can reverse a preference when the original scores are close. For each pair, the margin is the score of the benchmark-preferred sentence minus the score of its alternative. A positive margin indicates agreement with the benchmark. A change of sign indicates a reversal.

An association between reversals and small absolute baseline margins is partly a consequence of this decision rule. A point close to zero requires a smaller displacement to cross the boundary. Such an association can identify vulnerable decisions without identifying the representation of a particular suffix as the cause.

3.2. Tokenization and morphosyntax

With a fixed tokenizer, changing weight precision leaves the token sequence of a given sentence unchanged. Matching token identities across conditions is therefore a validity check. The empirical question is whether an existing characteristic of those sequences helps explain which decisions change. A suffix may be represented by different subword units in different contexts. The available records permit a test using a fragmentation proxy but do not identify morpheme boundaries or changes in internal representations.

4. Data and Task

We used four TurBLiMP categories: subject agreement, transitive argument structure, nominalization, and irregular forms. Each source file contained 1,000 pairs. We screened for empty sentences, identical alternatives, and exact duplicate pairs. No invalid rows or cross-category exact duplicates were found. Removing 338 within-category duplicates left a pool of 3,662 distinct pairs. Using Python’s random.Random(42) and a fixed category order, we sampled 250 pairs from each category. See Table 1.

Table 1. Sampling frame and selected pairs. Source: Author's calculations from the experimental records.

Category	Raw	Duplicates	Unique	Selected
Subject agreement	1000	66	934	250
Argument structure	1000	165	835	250
Nominalization	1000	55	945	250
Irregular forms	1000	52	948	250
Total	4000	338	3662	1000

A duplicate required an exact match of both sentences within a category.

The source revision recorded in the project manifest was 297de13fb7a0ce524fe32e8b175c6b5255d66960. The SHA-256 digest of the selected JSONL sample was bcb2c7f25475f8290d1511d742e1e9f8f515fd29df11dd3e3f64f2f53bcb4835. Resampling the local source files reproduced the content and order of the selected pairs. The digest documents file integrity. Linguistic label validity requires separate evidence.

The sample contained 1,000 unique pair identifiers and 1,000 distinct sentences on each side of the contrast. Equal category sizes make micro-accuracy identical to the unweighted mean of the four category accuracies. These weights describe the benchmark design rather than the prevalence of the phenomena in naturally occurring Turkish. Ellipsis, morphological tagging, and inflected-word generation were outside the evaluation scope.

Each sentence was scored separately. A decision was counted as correct when the benchmark-preferred sentence received the higher score. The task required neither free generation nor a verbal grammaticality judgment. The preferred and dispreferred sentences averaged 13.415 and 13.464 tokens, respectively, with ranges of 6–28 and 6–29. No truncation was required within the 4,096-token context window. Each configuration scored 26,879 sentence tokens, excluding the beginning-of-sequence context.

5. Methods

5.1. Model configurations

EuroLLM-1.7B is a multilingual decoder-only base model that includes Turkish (Martins et al., 2024; UTTER Project, n.d.). The local configurations shared the same architecture and tokenizer. The model has 24 layers, a hidden dimension of 2,048, 16 attention heads, and eight key/value heads. Q8 and Q4 used affine weight quantization with a group size of 64. No chat template, conversation history, demonstrations, or task-specific adaptation was used.

The reference is designated FP16 on the basis of the F16 tensors in the local weight file. The BF16 field in the source configuration describes a different aspect of the model. The quantized files contain

packed U32 weights and F16 auxiliary tensors. The Q4 designation therefore refers to weight representation and does not imply that all arithmetic was performed at four bits. Project records associate all configurations with the same base model. An incomplete archive of the exact upstream revision, conversion commands, and tensor inventory limits independent reconstruction.

5.2. Sentence scoring

Sentences were tokenized without automatically adding special tokens. One BOS token supplied the initial context. No EOS token was appended or scored. Each observed sentence token was scored conditional on the preceding context. Logits were cast to float32 before log-softmax, and the log probabilities of the observed tokens were averaged. This cast applied to score calculation rather than the entire model.

$$S_q(s) = \frac{1}{T}\sum_{t=1}^{T} \ln p_q(x_t \mid BOS, x_{<t}) \quad (1)$$

Here T is sentence length in tokens and q denotes the configuration. For pair i, the preferred and dispreferred sentences are denoted by s^+ and s^-. Accuracy is the proportion of strictly positive margins.

$$m_{q,i} = S_q(s^+) - S_q(s^-), A_q = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}(m_{q,i} > 0) \quad (2)$$

No exact ties occurred. Summed log probability was retained as a sensitivity analysis. Scores cover complete sentences, so a reversal cannot be attributed specifically to the probability of the critical suffix.

5.3. Paired inference

The statistical unit was the minimal pair. We compared each quantized configuration with FP16 using a two-sided exact McNemar test (McNemar, 1947). Let d count correct-to-incorrect transitions and u count incorrect-to-correct transitions. Under the null hypothesis, conditional on d + u discordant pairs, the number of transitions in one direction follows a binomial distribution with probability 0.5. We applied Holm adjustment (Holm, 1979) to the two primary comparisons at a significance level of .05. No time-stamped preregistration was documented.

Individual accuracy intervals used the 95% Clopper–Pearson procedure (Clopper & Pearson, 1934). For the paired difference, we calculated separate 97.5% Clopper–Pearson intervals for u/N and d/N. Subtracting the upper bound for degradation from the lower bound for improvement, and vice versa for the opposite endpoint, gives a conservative interval with at least 95% coverage under the independent-pair sampling model. These intervals apply to individual contrasts. Holm adjustment was applied to p values.

5.4. Dependence among related pairs

Distinct pairs may share a linguistic core. Within each category, we grouped pairs whose normalized alternatives matched after removing shared leading words from a specified list: ama, ancak, ardından, artık, aslında, ayrıca, belki, bugün, burada, evet, fakat, hadi, hemen, işte, sadece, sonra, ve, yalnız, yani, yine, çünkü, öncelikle, özellikle, and şimdi. The resulting 776 clusters approximate related examples. Their membership has not been validated against the original generation templates.

We used category-stratified bootstrap resampling (Efron, 1979) of complete pairs and, separately, of complete approximate clusters. Each procedure used 20,000 replicates for the overall accuracy contrast, with seeds 20261028 and 20261029, respectively. Categories retained equal weights. Dependence beyond that captured by the clustering rule may remain.

5.5. Exploratory fragmentation analysis

For each pair, the fragmentation proxy was the arithmetic mean of the two token-to-word ratios.

$$F_i = \frac{1}{2}\left(\frac{T^+}{W^+} + \frac{T^-}{W^-}\right) \quad (3)$$

T and W denote token and word counts. Words were counted as Unicode letter or digit sequences, preserving internal apostrophes and treating hyphens as boundaries. Standalone punctuation contributed to the token count but not the word count. The outcome of the logistic regression was any reversal between FP16 and Q4, regardless of direction.

The unadjusted model contained fragmentation alone. The main adjusted model also included absolute FP16 margin, mean word count, and category, with argument structure as the reference. All predictors, including category indicators, were standardized. We added a ridge penalty equal to λ/2 times the sum of squared slopes to the total negative log likelihood and left the intercept unpenalized. The main penalty was λ = 1, with 0.1 and 10 examined for sensitivity. No predictive tuning or held-out classifier evaluation was performed.

Pair and cluster resampling each used 2,000 replicates, with seeds 20261029 and 20261030. We refitted the scales and coefficients in each replicate and normalized category weights to their original sizes. Odds ratios were expressed per original-sample standard deviation of fragmentation. Percentile intervals describe uncertainty in the penalized estimates and do not correct regularization bias or multiplicity across exploratory models. Q8 had only three reversals and was not modeled separately.

6. Experiments

The experiment ran on an Apple M2 MacBook with 16 GiB of unified memory, macOS 26.4.1, Python 3.12.15, MLX 0.32.4, and MLX-LM 0.32.2. Batch size was one sentence. We processed 20 chunks of 50 pairs and recorded scoring times after loading and warm-up. The additional analyses used NumPy 2.5.4, SciPy 1.18.1, and Matplotlib 3.11.2.

The audit covered 3,000 pair–configuration records and 6,000 sentence records. Checks included identifiers, categories, token counts and digests, and score consistency. The maximum absolute discrepancy between the stored mean and the sum divided by token count was 4.24×10^{-7}. Recomputing the original statistics produced identical results.

Mean token log probability defined the primary analysis. Summed scores, category breakdowns, pilot exclusion, label sensitivity, margin distributions, and fragmentation models provided secondary or exploratory analyses. The pilot included five pairs per category drawn from the main sample. A complete rerun in separate processes reproduced every decision. It did not add independent observations. Background load was uncontrolled during the rerun, limiting performance comparisons.

7. Results

7.1. Overall accuracy and transitions

FP16, Q8, and Q4 produced 912, 915, and 884 correct decisions, respectively. Q8 retained every correct FP16 decision and corrected three errors. The small positive difference did not establish superiority. Equivalence or noninferiority would require a prespecified acceptable loss. See Table 2.

Table 2. Primary accuracy and paired contrasts against FP16. Source: Author's calculations from the experimental records.

Model	Correct / N	Accuracy % [95% CI]	Δ pp	Paired 95% CI, pp	Holm p
FP16	912/1000	91.2 [89.27, 92.88]	—	—	—
Q8	915/1000	91.5 [89.60, 93.15]	+0.3	[−0.390, 0.971]	.250
Q4	884/1000	88.4 [86.25, 90.32]	−2.8	[−5.151, −0.416]	.000469

pp = percentage points. Accuracy intervals are exact binomial intervals. Difference intervals use the conservative paired procedure.

Q4 reversed 56 decisions (5.6%). Forty-two were degradations and 14 were improvements, yielding a net loss of 28 correct decisions. FP16 and Q4 were both correct for 870 pairs and both incorrect for 74. The error count rose from 88 to 116, a relative increase of 31.8%. All three configurations agreed on 870 correct and 72 incorrect decisions.

For Q4, the category-stratified pair bootstrap interval was [−4.30, −1.40] pp and the approximate-cluster interval was [−4.45, −1.21] pp. For Q8, bootstrap lower bounds of zero reflect the absence of observed degradations: resampling cannot generate a transition type absent from the observed sample. The conservative interval is therefore the primary basis for interpretation.

7.2. Category differences

Subject agreement accounted for 18 of the 28 net lost decisions (64.3%). Argument structure and nominalization contributed four and seven net losses, respectively, while irregular forms contributed one net gain. See Table 3.

Table 3. Category accuracy and Q4 transitions (250 pairs per category). Source: Author's calculations from the experimental records.

Category	FP16 %	Q8 %	Q4 %	Q4 Δ pp	d / u
Subject agreement	88.4	88.8	81.2	−7.2	23 / 5
Argument structure	93.2	93.2	91.6	−1.6	6 / 2
Nominalization	84.0	84.8	81.2	−2.8	13 / 6
Irregular forms	99.2	99.2	99.6	+0.4	0 / 1

d and u denote degradations and improvements relative to FP16.

Across a separate exploratory Holm family of eight category comparisons, the adjusted p value for Q4 subject agreement was .00730. The other comparisons did not reach the .05 threshold. Differences in significance across categories do not establish an interaction. Irregular forms were already near ceiling at 99.2% under FP16, leaving only two errors available for improvement.

7.3. Baseline margins

Mean absolute margin shifts relative to FP16 were 0.00838 for Q8 and 0.10509 for Q4. Pearson correlations with FP16 margins were .99939 and .90822, respectively. Close agreement across the full distribution can coexist with sign changes near zero. Across successive absolute FP16 margin quartiles, Q4 reversed 46, 8, 1, and 1 decisions. The lowest quartile accounted for 82.1% of reversals. Its reversal rate was 18.4%, compared with 1.33% among the remaining 750 pairs. See Figure 1.

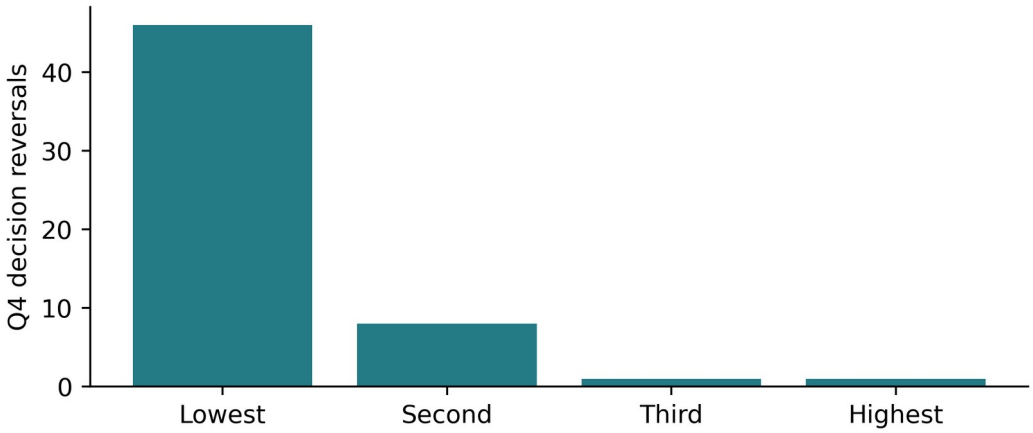


Figure 1. Q4 reversals by absolute FP16 margin quartile (250 pairs each). Source: Author's calculations from the experimental records.

7.4. Scoring and sample sensitivity

Under summed log probability, accuracy was 94.6% for FP16, 94.8% for Q8, and 90.7% for Q4. The Q4 contrast increased in magnitude to −3.9 pp. Switching between mean and summed scoring changed 60 FP16, 59 Q8, and 81 Q4 decisions. In the 626 pairs with equally long token sequences, both scoring rules yielded the same preference for every configuration. See Table 4.

Table 4. Sensitivity of the Q4–FP16 contrast. Source: Author's calculations from the experimental records.

Analysis	N	Δ pp	Paired 95% CI, pp	Holm p
Primary mean score	1000	−2.800	[−5.151, −0.416]	.000469
Summed score	1000	−3.900	[−6.176, −1.576]	3.04×10^{-7}
Pilot excluded	980	−2.755	[−5.137, −0.340]	.000711
Eight disputed pairs excluded	992	−2.520	[−4.747, −0.262]	.000940
Predicate screen excluded	983	−2.543	[−4.790, −0.265]	.000940
Subject agreement excluded	750	−1.333	[−3.656, 1.016]	.174317

Holm adjustments apply to the two FP16 comparisons within each analysis.

Excluding the pilot or potentially ambiguous nominalizations preserved the negative Q4 contrast. Excluding subject agreement reduced it to −1.333 pp (adjusted p = .174), illustrating sensitivity to category composition. These exclusions cannot establish label validity in the remaining sample.

7.5. Linguistic examples

A purposive review covered 20 pairs comprising 12 Q4 degradations, five Q4 improvements, and three Q8 improvements. The review was extended to all 58 unique pairs on which either quantized configuration differed from FP16. The preliminary judgments were AI-assisted. No independent blinded human annotation or inter-annotator agreement was obtained.

For subject_agreement:0532, the preferred sentence was “Ancak o bu projeye çok emek verdi.” The alternative substituted sen (‘you’) for o (‘he/she’) while retaining the third-person predicate verdi. The FP16, Q8, and Q4 margins were 0.301963, 0.304616, and −0.062743. For argument_structure_transitive:0850, “Hadi bu filme bayılıyoruz.” contrasts with “Hadi bu filmi bayılıyoruz.” The verb calls for the dative form filme under the intended reading. The corresponding margins were 0.039436, 0.041425, and −0.167631.

In nominalization:0962, the label favors “Ve şirketin yatırım kararlarını hızla almasına çoğu çalışan şaşırdı.” over the alternative with aldığına. Margins were −0.017708, −0.016909, and 0.236379. Event-oriented and fact-oriented interpretations may license both complements, so Q4’s agreement with the label is an ambiguous linguistic improvement. Of the 58 reviewed pairs, eight raised similar concerns, three were context-sensitive, and five required expert review. The selection does not support a corpus-wide estimate of labeling errors.

7.6. Subword fragmentation

The sample quartile thresholds were 1.857143, 2.083333, and 2.339286 tokens per word. Ties at a threshold remained in the lower group. The group sizes were 256, 247, 247, and 250. Reversal rates were 5.86%, 4.45%, 8.91%, and 3.20%, with no monotonic increase. The highest-minus-lowest contrast was −2.66 pp (approximate-cluster bootstrap 95% interval [−6.50, 1.41]). See Figure 2.

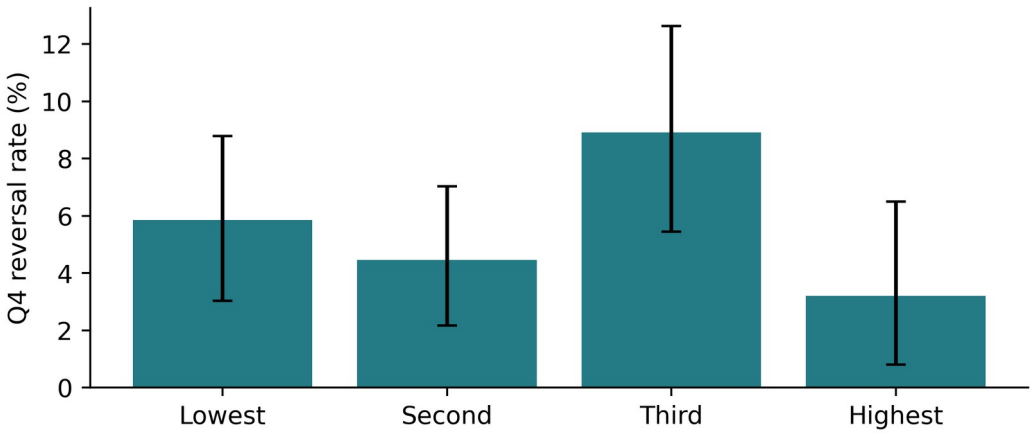


Figure 2. Q4 reversal rates by fragmentation quartile. Bars represent 15/256, 11/247, 22/247, and 8/250 pairs. Error bars show category-stratified approximate-cluster bootstrap 95% intervals. Source: Author’s calculations from the experimental records.

Table 5. Fragmentation and the odds of a Q4 reversal. Source: Author's calculations from the experimental records.

Model	OR per SD	Pair bootstrap 95% CI	Cluster bootstrap 95% CI
Unadjusted	.789	[.611, .982]	[.592, 1.027]
Margin, category, word length	.735	[.548, .932]	[.540, .946]
Margin, category, token length	.900	[.664, 1.193]	[.649, 1.230]

One SD corresponds to 0.3977 tokens per word. Intervals are exploratory and unadjusted for multiple comparisons.

Adjustment for within-pair fragmentation differences yielded an OR of .742 (cluster interval [.544, .952]). Using log absolute margin gave .737 [.546, .953]. Penalties of 0.1 and 10 produced point estimates of .716 and .817, without additional intervals. The negative adjusted association weakened when token count replaced word count. These results offer no evidence for a general protective effect of fragmentation. An odds ratio describes a change in odds, and the outcome includes both improvements and degradations. See Table 5.

7.7. Resource measurements

In the original run, scoring took 20.4% less time for Q8 and 22.6% less for Q4 relative to FP16. Measured peak MLX allocator memory fell by 63.8% for Q4, and weight-file size fell by 71.9%. See Table 6.

Table 6. Resource measurements in the original run. Source: Author's calculations from the experimental records.

Measure	FP16	Q8	Q4
Scoring time (s)	123.04	97.88	95.20
Peak MLX allocator memory (GB)	3.416	2.068	1.237
Peak process RSS (GB)	1.247	1.839	1.573
Weight files (GB)	3.314	1.761	0.932

GB denotes 10^9 bytes. Allocator memory, resident set size, and storage have distinct measurement scopes.

Timing includes tokenization, model computation, score extraction, record preparation, and cache clearing between pairs. Loading, warm-up, file writing, and pauses between chunks were excluded. The Q8–Q4 difference of 2.68 seconds represents one run. In the verification run, FP16, Q8, and Q4 took 127.08, 104.86, and 98.85 seconds, respectively, with identical decisions. The 20 chunks contain different sentences and cannot serve as 20 independent performance replicates. Energy consumption, temperature, memory pressure, and swap were not measured concurrently.

8. Discussion

The Q4 configuration reduced measured memory use at a cost in accuracy. Q8 remained close to the reference on this sample. The acceptable loss for deployment depends on the intended application, and no practical tolerance was specified before analysis. The observed trade-off consequently applies to the reported workload.

Subject agreement contributed disproportionately to the overall decline. This pattern cannot be attributed specifically to damaged person-suffix representations: lexical choices, template structure, and baseline difficulty vary across categories. No token- or layer-level intervention isolated an internal mechanism. Baseline margins help describe vulnerable examples, although they are not calibrated confidence probabilities or a validated routing criterion.

The fragmentation analysis addresses a different question from comparisons of alternative Turkish tokenizers (Toraman et al., 2023): here, the tokenizer is fixed and only weight precision changes. A word split into several tokens need not represent a difficult grammatical contrast. Punctuation contributes to the numerator of the proxy, and length enters both the ratio and the adjustment model. The token-length sensitivity interval includes the neutral value, which limits any interpretation of the negative slope as a robust linguistic effect.

Mean and summed scores yielded different absolute accuracy levels and different sizes of the Q4 deficit. Reporting both while retaining the original primary rule makes the effect of normalization visible. Comparisons with other minimal-pair studies should account for their scoring conventions.

9. Limitations

We evaluated one base model, one quantization scheme, one language, and four categories. Other model scales, monolingual Turkish models, and calibration-based methods may behave differently. No cross-language control permits attribution to Turkish or to agglutinative languages generally. Overlap between test sentences and pretraining data was not investigated.

Balanced category sampling does not represent natural usage or the entire benchmark. Exact deduplication leaves related templates, and the clustering heuristic captures only selected initial-word variants. Exploratory AI-assisted review does not substitute for blinded judgments by independent Turkish-language experts. Exclusion analyses address only the label issues identified.

Whole-sentence rankings do not measure inflection generation, lemmatization, or morpheme tagging. The fragmentation proxy lacks gold morpheme boundaries. The logistic model estimates a common linear log-odds slope without category interactions or nonlinear terms. Sparse events and regularization limit coefficient interpretation.

The pilot overlaps the main sample, the study was not preregistered, and multiplicity was not controlled across all exploratory analyses. A persistent public experiment archive and complete conversion manifest have not been specified for this version. Decision reproducibility on the same inputs is distinct from independent replication on new items. Timing comparisons were not based on repeated full runs with randomized configuration order.

10. Conclusion

Across 1,000 matched Turkish minimal pairs, Q8 differed from FP16 by +0.3 pp without establishing superiority. Q4 differed by −2.8 pp, a statistically significant decline after Holm adjustment. Its 56 reversals included changes in both directions, concentrated in low-margin items. Subject agreement accounted for most of the net decline.

The exploratory analysis did not support an increase in decision changes with greater fragmentation. The inverse conditional association depended on the definition of length. Alongside a 63.8% reduction in measured peak MLX allocator memory, these findings quantify an accuracy–memory trade-off for this model, sample, and scoring procedure. Blinded expert review, verified generation templates, and replication on another checkpoint would test the limits of the result.

Research Transparency

Data and code availability. The original TurBLiMP data are available through the repository identified in the benchmark publication (Başar et al., 2025). Numerical summaries are recorded in main_statistics.json, extended_statistics.json, and fragmentation_statistics.json. The selected example reviews are recorded in manual_review_20_annotated.csv. This version provides no persistent public archive for the experimental code, selected sample, or conversion records.

Use of artificial intelligence. AI tools assisted with manuscript preparation and preliminary linguistic review. The quantitative findings derive from experimental records. No new data were collected from human participants.

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